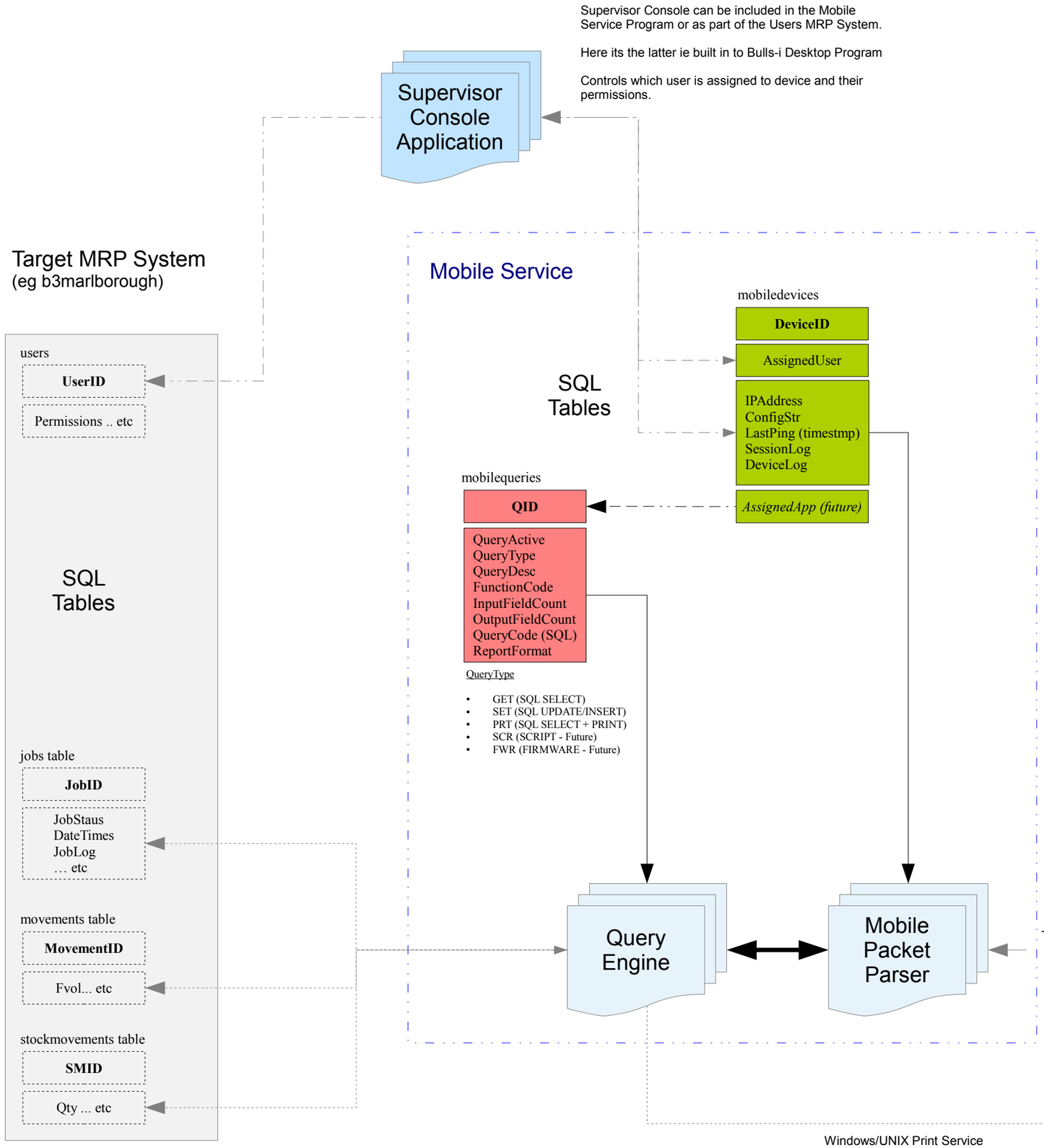
Indevin Bulls-i Mobile System



Simple TCP Packet Message Format

Encoding ASCII. TAB-DELIMITED-DATA is ASCII chars >= 0x20 and <=0x7f (reject other chars)

Special ASCII Chars are SOH=0x01, STX=0x02, ETX=0x03, EOT=0x04, ACK=0x06, NACK=0x15

Direction	Message Details
MOB->SVR	SOH **FnCode** STX **TAB-DELIMITED-DATA** ETX **HWRevByte** **SWRevByte** EOT
Respond OK	SOH **ACK** STX **TAB-DELIMITED-DATA** ETX EOT
Respond ERR	SOH **NACK** STX "**ErrorCode**", "**Error Message**" ETX EOT

Note All queries require the mobile device to be assigned to a user, if not assigned then retuns unnassigned error
Actual SQL Queries are stored in query table using function to lookup SQL and fields substituted real-time to form query

FnCode

GET (SELECT)

0x10	MOB->SVR	GetJob, Send Respond OK:	Field1 = JobID Field1 = JobStatusByte, Field2 = JobStatusString, Field3 = JobDescriptionString
0x11	MOB->SVR	GetAssetVC, Send: Respond OK:	Field1 = AssetID Field1 = AssetName, field2 = AssetStatus, Field3 = CurrVolume, Field4 = BatchCode, field5 = CurDip, Field6 = StatusCode
0x12	MOB->SVR	GetAssetPosn, Send: Respond OK:	Field1 = AssetID Field1 = AssetName, field2 = AssetStatus, Field3 = Posn, Field4 = Locn
0x13	MOB->SVR	GetAssetSCADA Respond OK	Field1 = AssetID Field1 = CurTemp, field2 = CurSetpoint, Field3 = Valve Status, field4 = agi Status
0x14	MOB->SVR	GetMovement, Send: Respond OK	Field1 = JobId, Field2 = AssetID Field1 = MovementID, field2 = AssetName, Field3 = AssetStatus, Field4 = EstiDip, Field5 = iDip, Field6 = EstfDip, Field7 = fDip, Field8 = fVol, Field9 = Num Additions, field10 = PadlockCode
0x15	MOB->SVR	GetAdditions, Send: Respond OK	Field1 = MovementID Field1 = SMID1Desc, Field2 = SMID2Desc, Field3 = SMID3Desc, Field4 = SMID4Desc, field5 = SMID5Description
0x16	MOB->SVR	GetSMID, Send: Respond OK	Field1 = SMID (from product stock label barcode ID scan) Field1 = ProductName, Field2 = AddQty (Incl UOM), Field3 = Addn Status

SET (EXECUTE) *All SET functions respond with generic OK (ACK) no fields set, wifi password is used as auth/confirmation*

0x81	MOB->SVR	SetJobInProgress, Send: Field1 = JobID, Field2 = WiFi Password
0x82	MOB->SVR	SetJobComplete, Send: Field1 = JobID, Field2 = WiFi Password
0x92	MOB->SVR	UnquarantineAsset, Send Field1 = AssetID, Field2 = WiFi Password
0x93	MOB->SVR	SetAssetLocn, Send: Field1 = AssetID, Field2 = NewLocn, Field3 = NewPosn, Field4 = WiFi Password
0x94	MOB->SVR	SetAssetCleaned, Send: Field1 = AssetID, Field2 = WiFi Password
0x95	MOB->SVR	SetAssetQuarantine, Send: Field1 = AssetID, Field2 = WiFi Password
0xA0	MOB->SVR	SetInitialDip, Send Field1 = MovementID, Field2 = iDip, Field3 = WiFi Password
0xA1	MOB->SVR	SetFinalDip, Send: Field1 = MovementID, Field2 = iDip, Field3 = WiFi Password
0xB0	MOB->SVR	SetAdditionComplete, Send: Field1 = SMIID, Field2 = WiFi Password

PRT (Print Report/label)

0xe0	MOB->SVR	Print Asset label, Send: Field1 = AssetID, Field2 = PRINTERNAME (Optional)
0xe1	MOB->SVR	Print Tank VC Label, Send: Field1 = AssetID, Field2 = PRINTERNAME (Optional)

System is stateless, ie each packet should be a self contained message not relying on information from previous packet.